Functii C (13 probleme rezolvate cu functii)

```
/*************************************************************************/
    1. //Ce scrie programul de mai jos?
#include <stdio.h>
void f1(int *a, int *b){    int *aux=a; a=b; b=aux; }

int main(){
   int a=5, b=7;
   f1(&a, &b);
   printf("a=%d b=%d", a, b);
   return 0;
}

/*************************************************************************/
    2. //Ce scrie programul urmator?
#include <stdio.h>
int f2(int a, int b){
   a=a+1; b++;
   printf("%d %d\n", a, b);
   return (a+b);
}

int main(){
   int a=5, b=7;
   printf("s=%d a=%d b=%d", f2(a, b), a, b);
   return 0;
}

/*************************************************************************/
    3. //Ce scrie programul urmator?

#include <stdio.h>
int s(int *a, int b)
{
   *a=*a+1; b=b+1;
   printf("%d %d\n", *a, b);
   return *a+b;
}

int main(){
   int a = 5, b = 7, c = s(&a, b);
   printf("s = %d a = %d b = %d", c, a, b);
   return 0;
}
```

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/*************************************************************************/
    4.  //Ce scrie programul urmator?
#include <stdio.h>
void s(int *a, int *b)
{
   *a = *a + 1; *b = *b + 1;
   printf("%d %d\n", *a, *b);
}

int main()
{
   int a = 5, b = 7;
   s(&a, &b);
   printf("a = %d b = %d", a, b);
   return 0;
}

/*************************************************************************/
    5.  //Ce scrie programul urmator?
#include <stdio.h>
#include<math.h>
int F(int n)
{
   int s=1;
   for(int i=2; i<sqrt(n*n);i++)
   if(!(n%i)){s+=i; printf("%d ", i);}
   return s;
}

int main()
{
   int x=12;
   int c = F(x);
   printf("\ns = %d ", c);

   return 0;
}
```

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/*************************************************************************/
     6.   //Ce scrie urmatorul program pentru x = 10? Dar pentru x = 40?
#include <stdio.h>
#include<math.h>
int p(int n)
{
   int s = 1;
   for(int i = 2; i < sqrt(n*n); i++)
   if(!(n % i))s += i;
   return s;
}
void scrie_perechi(int n)
{
  int ok = 0;
  for (int k = 1; k < n; k++)
       for(int j = k + 1; j <= n; j++)
          if(p(k) == j)
          {
             printf("(%d,%d)\t", k, j);
             ok++;
          }
   if(ok == 0)
      printf("Nu exista");
}
int main()
{
   int x; scanf("%d", &x);
   scrie_perechi(x);
   return 0;
}
```

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/*************************************************************************/
    7.  //Ce scrie urmatorul program pentru 1562? Dar pentru 33?
#include <stdio.h>
#include<math.h>
void cx(int n, int *ma, int *mi)
{
   *ma=n%10;
   *mi=n%10;
   n/=10;
   while(n)
   {
      int c=n%10; n=n/10;
      if(*ma<c) *ma=c;
      if(*mi>c) *mi=c;
   }

}
int main()
{
   int x, m1, m2; scanf("%d", &x);
   cx(x, &m1, &m2);
   printf("%d %d", m1, m2);
   return 0;
}
```

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/*************************************************************************/
    8.  //Ce scrie urmatorul program pentru 7? Dar pentru 34?
#include <stdio.h>
#include<math.h>
void f1(int n, int *a, int *b) {
   *a = n % 10;
   *b = n % 10;
   n /= 10;
   while(n) {
      int c = n % 10; n = n / 10;
      if(*a < c) *a = c;
      if(*b > c) *b = c;
   }

}
void f2(int x)
{
  int m1, m2;
  for (int k = 0; k <= x; k++)
    {
       f1(k, &m1, &m2);
       if (m1 == m2)
       {
          printf("%d ",k);
       }
    }
}
int main()
{
   int n, x, y;
   scanf("%d", &n);
   f2(n);
   return 0;
}
```

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/*************************************************************************/
    9.  // Ce problema rezolva programul urmator?
#include <stdio.h>
#include <stdlib.h>
void f1(int *n, int v[])
{
        FILE *f;
        f = fopen("date.in", "r");
        if(f == NULL)
           {printf("Nu s-a putut deschide fisierul."); exit(1);}
        fscanf(f, "%d", n); // atentie!
        for(int i = 0; i < *n; i++)
                fscanf(f,"%d", &v[i]);
        if (fclose(f) != 0) printf("Probleme la inchiderea fisierului.");

}
void f2(int n, int v[])
{

        for (int k = 0; k < n; k++) printf("%d ", v[k]);
        printf("\n");
}
int main()
{
        int n, v[20]; f1(&n, v); f2(n, v); // atentie la apel!
        return 0;
}
```

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/*************************************************************************/
     10. // Ce problema rezolva urmatorul program?
#include <stdio.h>
#include <stdlib.h>
void citire(int *n, int v[]) {
   FILE *f; f = fopen("date.in", "r");
   if(f == NULL)    { printf("Nu s-a putut deschide fisierul.");  exit(1); }
   fscanf(f, "%d", n);
   for(int i = 0; i < *n; i++)   fscanf(f, "%d", &v[i]);
   if (fclose(f) != 0)  printf("Probleme la inchiderea fisierului.");
}

void scrie(int n, int v[])
{
   for (int k = 0; k < n; k++)
   {    printf("%d ", v[k]);   }
   printf("\n");
}

int val(int t, int q, int v[])
{
   int m = v[t];
   while (t <= q)
   {
      if(v[t] > m)
         m = v[t];
      t++;
   }
   return m;
}

int main()
{
   int n, v[20]; citire(&n, v); scrie(n, v);
   int i = 6, j = 9; // unde n > j!
   printf("\n%d ", val(0, n-1, v));
   printf("\n%d ", val(i , j, v));
   return 0;
}
```

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/*************************************************************************/
    11. // Scrieti un posibil enunt al problemei pe care o rezolva programul urmator:
#include <stdio.h>
#include <stdlib.h>
void citire(int *n, int v[])  {
   FILE *f;
   f = fopen("date.in", "r");
   if(f == NULL)
   { printf("Nu s-a putut deschide fisierul."); exit(1); }
   fscanf(f, "%d ", n);     for(int i = 0; i< *n; i++)  fscanf(f,"%d", &v[i]);
   if (fclose(f) != 0)   printf("Probleme la inchiderea fisierului.");
}

int val_max(int t, int q, int v[])  {
   int Max = v[t];
   while (t <= q)
   {
      if(v[t] > Max)  Max = v[t];
      t++;
   }
   return Max;
}

int f(int n, int v[])  {
   int i = 0;
   do{
      int x = val_max(i, i, v);
      int y = val_max(i+1,i+1, v);
      if(x != y) return 0;
      i++;
   }while(i < n-1);
   return 1;
}

int main()  {
   int n, v[20];
   citire(&n,v);
   printf("valoarea maxima = %d ", val_max(0, n-1, v));
   printf("\n%s ", (f(n, v))?"Da":"Nu");
   return 0;
}
```

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/*************************************************************************/
    12. /*Ce problema rezolva programul? Dati doua exemple concludente de date de intrare si
        scrieti datele de iesire corespunzatoare. */
#include <stdio.h>
#include <stdlib.h>

void citire(int *n, int v[])
{
   FILE *f;
   f = fopen("date.in", "r");
   if(f == NULL)
   { printf("Nu s-a putut deschide fisierul."); exit(1); }

   fscanf(f, "%d ",n);
   for(int i = 0; i< *n; i++)
      fscanf(f,"%d", &v[i]);

   if (fclose(f) != 0)
      printf("Probleme la inchiderea fisierului.");
}

int val_max(int t, int q, int v[])
{
   int Max = v[t];
   while (t <= q)
   {
      if(v[t] > Max)
         Max = v[t];
      t++;
   }
   return Max;
}

int f(int x, int k)
{
   return x == k;
}

int cautare(int n, int v[], int k)
{
   int gasit = 0, i = 0;
   int (*pf)(int, int);// pointer la functie
   pf = f;
   while(!(gasit) && i<n)
```

```
    {
        if((*pf)(v[i], k))
            return i;
        i++;
    }
    return -1;
}

int egale(int n, int v[])
{
    int i = 0;
    int (*pf)(int, int);// pointer la functie
    pf = f;
    do{
        int x = val_max(i, i, v);
        int y = val_max(i+1, i+1, v);
        if (!((*pf)(x,y))) //(x!=y)
            return 0;
        i++;
    }while(i < n-1);
    return 1;
}

int main()
{
    int n, v[20];
    citire(&n,v);
    int k = 3;
    printf("Pozitia = %d\n", cautare(n, v, k));
    printf("Valoarea maxima in vector= %d\n", val_max(0, n-1, v));
    printf("%s\n",(egale(n,v))?"Da":"Nu");
    return 0;
}
```

```
/*************************************************************************/
    13. //Descrieti functiile din exemplul urmator. Ce problema rezolva programul?
#include <stdio.h>
#include <stdlib.h>

void citire(int *n, int v[])
{
   FILE *f;
   char *s; scanf("%s", s);
   f = fopen(s, "r");
   if(f == NULL)
   { printf("Nu s-a putut deschide fisierul."); exit(1); }

   fscanf(f, "%d ",n);
   for(int i = 0; i< *n; i++)
      fscanf(f,"%d", &v[i]);

   if (fclose(f) != 0)
      printf("Probleme la inchiderea fisierului.");
}

void scrie(int n, int v[])
{
   FILE *g;
   g = fopen("date.out", "w");
   if(g == NULL)
   { printf("Nu s-a putut deschide fisierul."); return;}

   for (int k = 0; k < n; k++)
   {
      fprintf(g, "%d ", v[k]);
   }
   fprintf(g, "\n");

   if (fclose(g) != 0)
      printf("Probleme la inchiderea fisierului.");
}

void schimbare(int n, int v[])
{
   int i = 0, k, nr = 0;
   for (int k = 0; k < n; k++)
   {
      if(v[k] == 0) nr++;
```

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    }
    printf("nr valori nule = %d\n", nr);
    while(i < n-2 && (nr))
    {
        if(v[i] == 0)
            {
                for(k = i; k < n-1; k++)
                    v[k] = v[k+1];
                v[k] = 0;
                nr--;
            }
        else i++;
    }

}

int main()
{
    int n, v[20];
    citire(&n, v);

    for (int k = 0; k < n; k++)
    {
        printf("%d ", v[k]);
    }
    printf("\n");
    schimbare(n, v);
    scrie(n, v);
    return 0;
}
```